CPE 314 — Software Design & Software Design Laboratory

Project Proposal

UC-B Campus Marketplace with Reviews

Table 1: Group information

Group Name / No.	Group 1 / Team 01 - BSCPE 3B - EDP 69323
Members	John Benedict Alaiza
	Acexion B. Mahigugmaon
	Glyde V. Arreglado
	Tristan Dominic Urot
	Joshua Brian S. Enovero
	Emanuelle De Paz
Repository URL	github.com/XionAce/cpe314-69323-capstone-team-01
Theme	Campus Marketplace with Reviews
API You Will Use	FakeStoreAPI — https://fakestoreapi.com
Project title	UC-B Campus Marketplace with Reviews
Document Version / Date	V 0.2 / 8 November 2026

1. One-Paragraph Pitch

UC-B Campus Marketplace with Reviews is a local Java desktop app for students, faculty and school groups to post and browse campus goods. Textbook, Electronics and Furniture listings have their own condition rules. Members can search nested categories, join waitlists and review listings. The app imports sales history from JSON and displays average prices and ratings by category. The prototype uses demo members, with a login page planned.

2. Entity-Relationship Sketch

The revised ERD shows all ten application tables and all their attributes. Listings is repeated across three views to keep the diagram readable. PK = primary key, FK = foreign key, UK = unique key. Complete SQL and constraints are in Section 4 of the project design document.

Read 1:M as one parent to many children. A parent can have no children. M:N (also called M:M) means many-to-many. Members and Listings have this relationship through BuyerInterests for waitlists and through Reviews for ratings. Each connecting table has two 1:M links. Each individual listing subtype uses 1:1; only the matching subtype has a row.

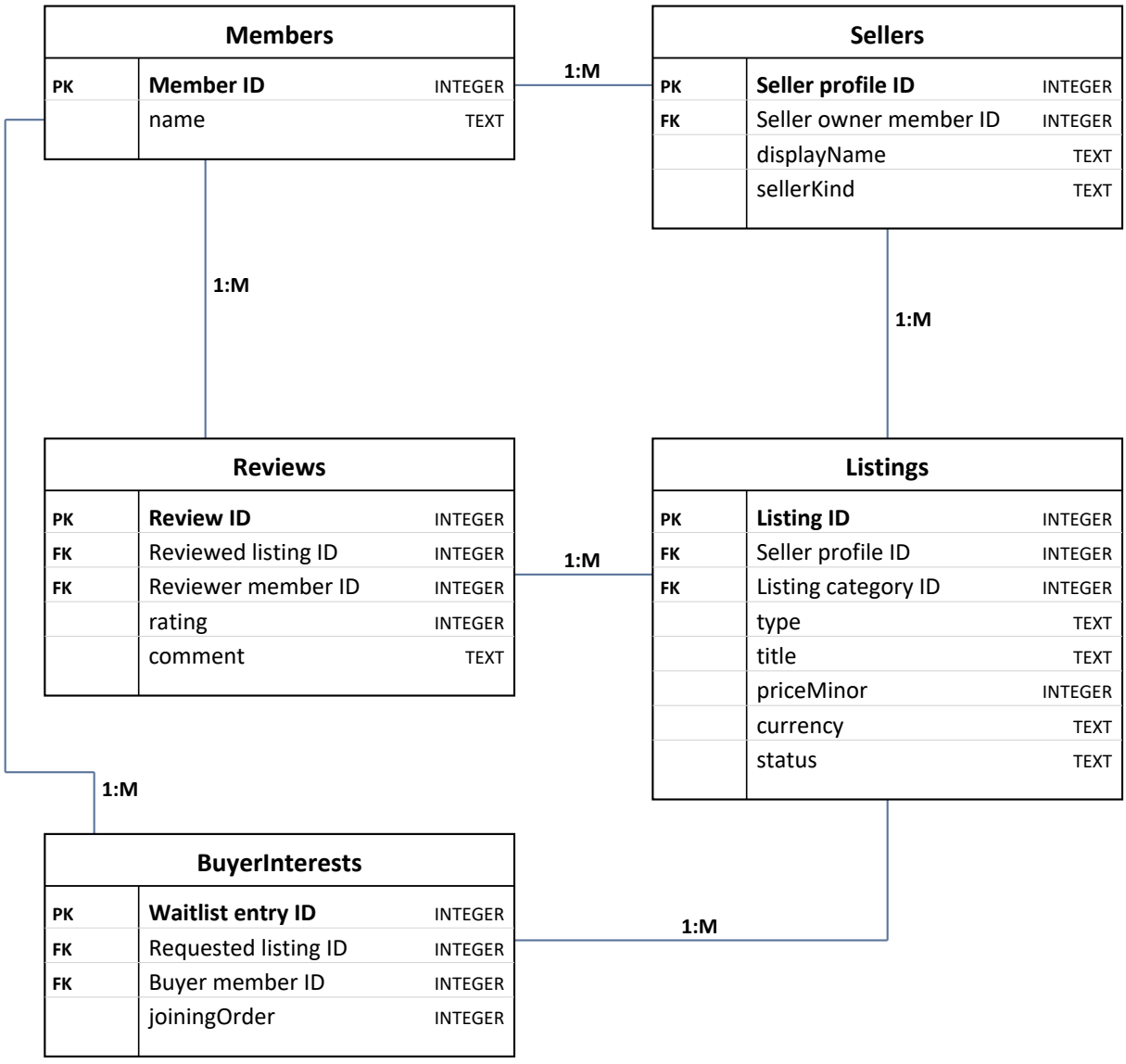


Figure 1: Members, sellers, listings, reviews and buyer interests

Categories and imported sales

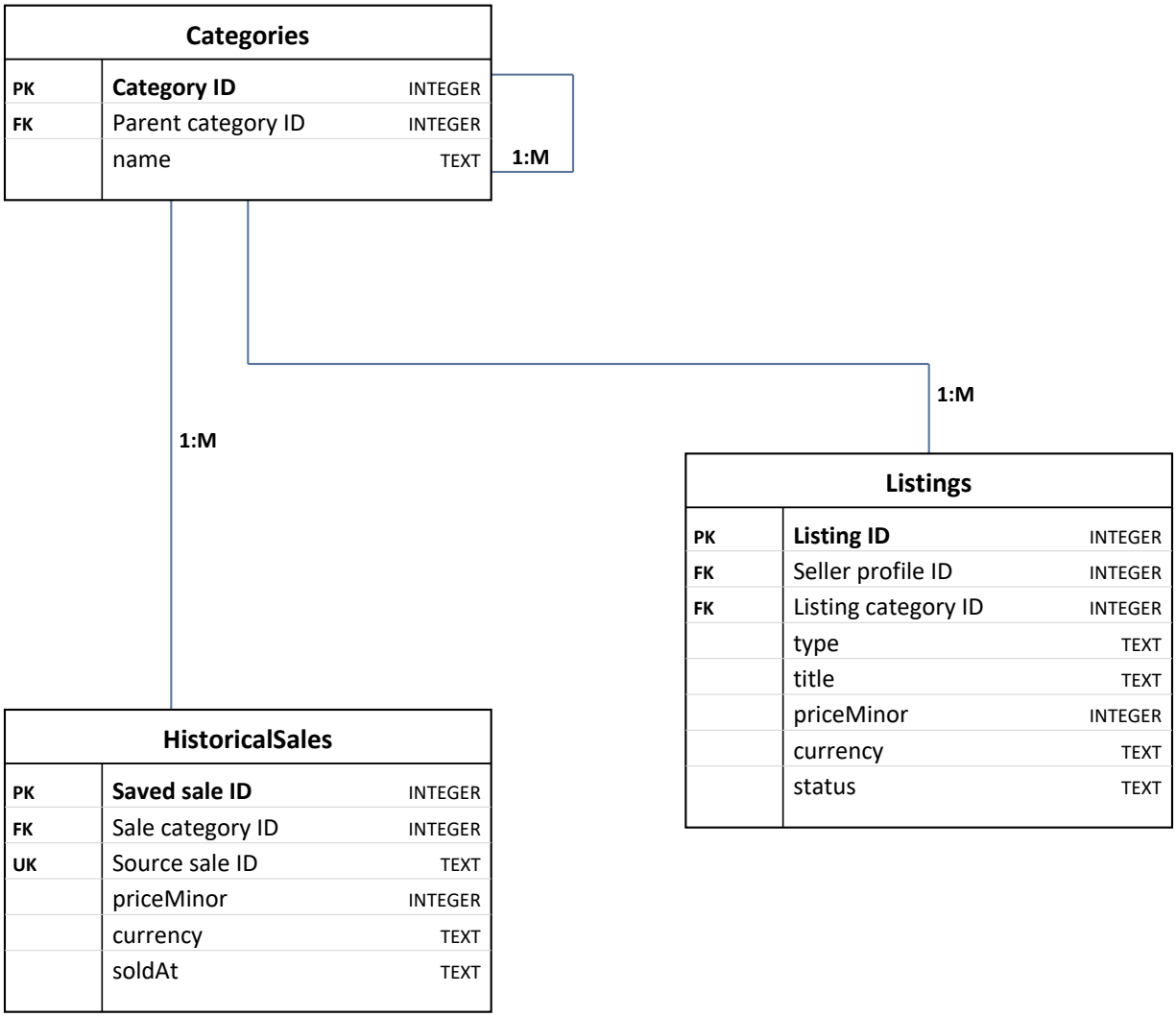


Figure 2: Categories, listings and historical sales

Each category has zero or one parent and zero or more children. Current listings and historical sales each belong to one category.

Condition details

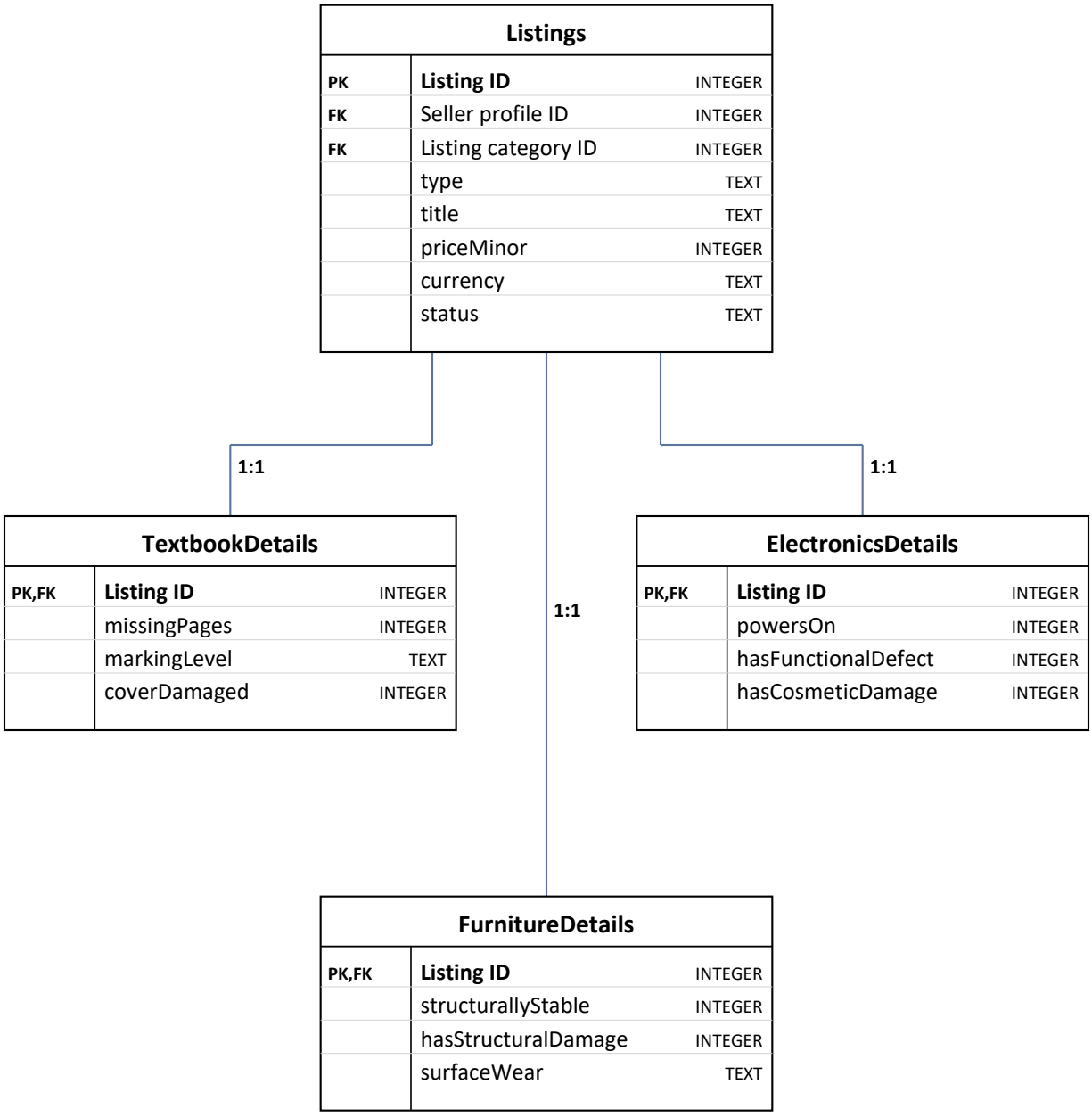


Figure 3: Type-specific listing details

Each listing has exactly one matching subtype-details row overall. listingId is both the PK and FK in each detail table. Transactions and Java validation enforce the matching-type rule.

Whose ID is it?

A primary key (PK) identifies one row in its own table. A foreign key (FK) identifies a related row. In the diagrams, Member ID means Members.id, Seller profile ID means Sellers.id, and Listing ID means Listings.id. The SQLite name id always belongs to the table where it appears.

Table 2: IDs used to connect marketplace records

SQLite field	What it identifies
Sellers.memberId	The member who owns the seller profile. Links to Members.id.
Listings.sellerId	The seller profile that posted this item. Links to Sellers.id.
Listings.categoryId	The item's category. Links to Categories.id.
Categories.parentId	The parent category. Links to Categories.id; empty for a root category.
Reviews.listingId	The listing being reviewed. Links to Listings.id.
Reviews.reviewerId	The member who wrote the review. Links to Members.id.
BuyerInterests.listingId	The listing whose waitlist the buyer joined. Links to Listings.id.
BuyerInterests.buyerId	The member who joined the waitlist. Links to Members.id.
HistoricalSales.categoryId	The category of the imported sale. Links to Categories.id.
TextbookDetails.listingId, ElectronicsDetails.listingId, FurnitureDetails.listingId	In each table, this is both PK and FK. It identifies the listing whose condition is described and links to Listings.id.

Example: Review ID 7, Reviewed listing ID 12 and Reviewer member ID 3 mean that member 3 wrote review 7 about listing 12.

HistoricalSales.id identifies the locally saved sale. HistoricalSales.sourceRecordId is the unique ID supplied by the JSON file to recognize a repeat import.

3. Feature Checklist

Table 3: Features and course requirements

Planned Feature	Requirement(s) It Satisfies
Abstract Listing with three condition-grading subclasses	Object-oriented design and polymorphism
FIFO waitlist, LIFO recent history and recursive category search	Data structures and recursion
JUnit normal and edge-case checks	Automated testing
Related SQLite tables and PreparedStatement persistence	Database modeling and JDBC
Live FakeStoreAPI catalog suggestions	API consumption
JavaFX controls and background JSON/API operations	Event-driven GUI and concurrency
Category means, counts and labeled bar charts	Data mining and visualization

4. Task Split

Table 4: Team responsibilities

Member and role	Primary Responsibility
John Benedict Alaiza - Backend Developer (Listing Models)	Listing classes, condition rules and validation.
Acexion B. Mahigugmaon - Project Leader / Integration	Connect the components, maintain documents and coordinate fixes.
Glyde V. Arreglado - Database Admin	SQLite tables, saved records and summary queries.
Tristan Dominic Urot - UI/GUI Developer	JavaFX controls, event handlers and charts.
Joshua Brian S. Enovero - Backend & API Developer	Catalog requests, JSON parsing and import.
Emanuelle De Paz - QA & Testing	Test cases, expected calculations and error checks.

5. Public API You Plan to Use

FakeStoreAPI GET https://fakestoreapi.com/products returns product JSON without an API key. Electronics title matching supplies category and catalog reference-price suggestions. API currency is unspecified; the member enters the final PHP price. Unsupported or unmatched items keep manual entry available. See API documentation.

6. Risks / Open Questions

1. If FakeStoreAPI matches a new electronics product but gives no currency, how will we prevent a misleading PHP resale-price suggestion?

2. If a listing has several campus reviews, how will we stop its price from being counted several times in the category price average?

3. If the same sales JSON file is imported twice, or cancellation occurs after some rows are processed, how will HistoricalSales avoid duplicate or partially saved sales?

4. If a Listings row loses its TextbookDetails, ElectronicsDetails or FurnitureDetails record, how will its owner restore the actual condition without replacing the listing or its reviews?